OPERATING SYSTEMS – ASSIGNMENT 1

TASK 4 – Evaluate the Services an Operating System Must Provide (4 Marks)
CO1–CO5 | Bloom's Level: K5 – Evaluate

Introduction

An operating system provides services that help users and programs use computer hardware safely and efficiently. The hospital should enable the services required by each device and avoid unnecessary services on small or safety-critical devices.

1. Important operating-system services

User interface: Allows users to interact with the system. It is needed on tablets and workstations; a bedside monitor may use a simple display only.
Program execution: Loads, runs and terminates applications. It is required in all environments.
I/O operations: Controls sensors, displays, keyboards, storage and network devices. This is essential for monitors and imaging equipment.
File-system management: Creates, reads, writes and organises files. It is important for servers and workstations; a monitor may use limited local storage.
Communication: Supports data exchange between processes and computers. It is needed for tablets, servers and partner-hospital links.
Error detection: Detects hardware and software errors. It is critical for monitors, database and imaging systems.
Resource allocation: Shares CPU, memory and I/O devices among programs. It is important on multi-user servers and workstations.
Accounting: Records resource usage and activities. It helps server monitoring and auditing.
Protection and security: Controls access and protects patient information. It is essential throughout the hospital.

2. Services required in each environment

Bedside monitor: Program execution, sensor I/O, error detection, basic communication, resource control and strong device protection. A full desktop interface and unnecessary background services can be excluded.
Nurses' tablet: Touch user interface, program execution, wireless communication, file access, power management, error handling and secure authentication.
Doctors' workstation: Full user interface, application execution, file and network services, resource allocation, audit/accounting and security.
Database server: File and storage management, communication, process execution, error detection, resource allocation, auditing, access control, backup and recovery services.
MRI imaging server: Program execution, high-speed I/O, file management, communication, multiprocessor resource allocation, error detection and access control.
Partner-hospital link: Secure communication, authentication, encryption, error checking, logging and controlled data exchange.

3. Services that can be reduced on bedside monitors

The monitor can use a simple user interface rather than a full desktop environment. Unnecessary multimedia applications, general-purpose printing services, unrelated background programs and unused network services can be removed. The device should retain sensor input, alarm output, timing, essential communication, error detection, secure updates and access protection. Removing unnecessary services reduces resource use and the attack surface.

4. Essential services for the database server

The database server needs reliable file and storage management, process and memory management, network communication, error detection, resource allocation, user authentication, authorisation, audit logs, backup and recovery. It must support concurrent access while preventing unauthorised changes or disclosure of patient records.

5. Protecting one patient's records from another user's programs

The OS should authenticate each user and assign permissions according to the user's role. Process isolation and memory protection prevent one program from reading another program's memory. File permissions and database access controls restrict access to authorised records. Encryption protects records during storage and transfer. Audit logs record who accessed or changed information. These controls should follow the hospital's privacy and security policies.

6. Alternative concept: full desktop OS on tablets

A full desktop OS may include many services and background processes that a tablet does not need. These services use extra memory and storage, leaving fewer resources for clinical applications. More services can increase start-up time because additional programs and drivers must load. Background activity and a larger system workload may consume more battery. A reduced handheld OS is therefore better suited to tablets because it focuses on touch interaction, wireless access and power management.

Conclusion

The hospital should enable essential OS services for every environment and remove unnecessary services where possible. Strong protection and security must remain enabled, especially for patient records and monitoring devices.

Prepared from the corresponding task in the uploaded CSA04 Operating Systems Assignment 1 document. Add your name, register number, required screenshots and your own measured evidence wherever requested before submission.